

Other Types of Equations

absolute value equations, e.g. $3|x+1|-2=-11$

$$\Rightarrow 3|x+1| = -9$$

$$\Rightarrow |x+1| = -3$$



No solutions
absolute value
is never negative

If instead
 $|x+1|=3$
 $x+1=3$ or -3
 $x=2$ or -4

e.g. $|x-2|+5=5$

$$\Rightarrow |x-2| = 0$$

$$\Rightarrow x-2=0$$

$$\Rightarrow \boxed{x=2}$$

rational exponent equations, e.g. $y^{3/2} = 5y$

Rearrange (get 0 on side)

$$y^{3/2} - 5y^1 = 0$$

$$y^1 (y^{1/2} - 5) = 0$$

either $y^1 = 0$

$$\downarrow$$
$$y = 0$$

OR $y^{1/2} - 5 = 0$

$$\downarrow$$
$$y^{1/2} = 5$$
$$(y^{1/2})^2 = 5^2$$
$$y^1 = 25$$
$$y = 25$$

e.g. $y^{4/3} = -3y$

$$y^{4/3} + 3y = 0$$

$$y(y^{1/3} + 3) = 0$$

either $y = 0$

$$\downarrow$$
$$y = 0$$

OR

$$y^{1/3} + 3 = 0$$

$$\downarrow$$
$$y^{1/3} = -3$$
$$(y^{1/3})^3 = (-3)^3$$
$$y = -27$$

quadratic type equations, e.g. $2x^4 - 10x^2 + 8 = 0$

$$2x^4 - 10x^2 + 8 = 0$$

$$\Rightarrow 2(x^4 - 5x^2 + 4) = 0$$

$$x^4 = (x^2)^2$$

$$x^4 = u^2$$

$$u = x^2$$

SUBSTITUTION

$$\Rightarrow \cancel{2}(u^2 - 5u + 4) = \cancel{2}0$$

$$\Rightarrow u^2 - 5u + 4 = 0$$

$$\Rightarrow u^2 - 1u - 4u + 4 = 0$$

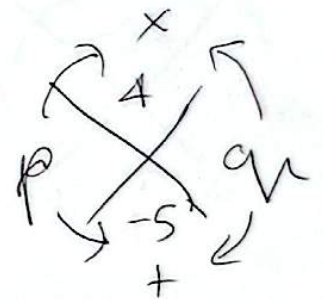
$$\Rightarrow \underline{u(u-1)} - 4\underline{(u-1)} = 0$$

$$\Rightarrow (u-4)(u-1) = 0$$

either $u-4=0$ OR $u-1=0$

$$u = 4 \text{ OR } u = 1$$

$$u = x^2 \Rightarrow \begin{cases} x = 1, -1, 2, \text{ OR } -2 \\ x = \pm 1 \text{ OR } \pm 2 \end{cases}$$



$$\begin{array}{r|l} 1 & 4 \\ \hline -1 & -4 \end{array}$$

$$-g. \quad 3x^{2/3} + 4x^{1/3} - 4 = 0$$

$$u = x^{1/3}$$

$$3(x^{1/3})^2 + 4x^{1/3} - 4 = 0$$

$$3u^2 + 4u - 4 = 0$$

$$3u^2 + 6u - 2u - 4 = 0$$

$$3u(u+2) - 2(u+2) = 0$$

$$(3u-2)(u+2) = 0$$

$$\downarrow \quad \downarrow$$

$$u = \frac{2}{3} \text{ or } u = -2$$

$$\begin{array}{cc} x & \\ \swarrow & \searrow \\ p & q \\ \swarrow & \searrow \\ & + \end{array}$$

$$p = 6, q = -2$$

$$u = x^{1/3} \Rightarrow x = u^3$$

$$x = \left(\frac{2}{3}\right)^3 = \frac{8}{27}$$

$$\text{or } x = (-2)^3 = -8$$

$$g. \left(\frac{x}{x-2} \right)^2 - 2 \frac{x}{x-2} - 15 = 0$$

$$u = \frac{x}{x-2}$$

$$u^2 - 2u - 15 = 0$$

$$(u-5)(u+3) = 0 \quad \leftarrow \begin{array}{l} \text{p\&q} \\ \text{method} \end{array}$$

$\downarrow$ $\downarrow$
 $u=5$ or $u=-3$

either $\frac{x}{x-2} = u = 5$ OR

$$\frac{x}{x-2} = 5$$

$$\Rightarrow x = 5(x-2)$$

$$\Rightarrow -4x = -10$$

$$\Rightarrow x = \frac{5}{2}$$

$$\frac{x}{x-2} = u = -3$$

$$\frac{x}{x-2} = -3$$

$$\Rightarrow x = -3(x-2)$$

$$\Rightarrow 4x = 6$$

$$\Rightarrow x = \frac{3}{2}$$

ical equations, e.g. $\sqrt[3]{x^2-1} = 2$

$$\Rightarrow (\sqrt[3]{x^2-1})^3 = 2^3$$

$$\Rightarrow x^2 - 1 = 2^3$$

$$\Rightarrow x^2 - 1 = 8$$

$$\Rightarrow x^2 = 9$$

$$\Rightarrow x = \pm\sqrt{9}$$

$$x = \pm 3$$

$$\sqrt{2x-3} - \sqrt{x+7} + 2 = 0$$

$$\Rightarrow (\sqrt{2x-3} + 2)^2 = (\sqrt{x+7})^2$$

$$\Rightarrow (2x-3) + 2 \cdot 2 \cdot \sqrt{2x-3} + 4 = x+7$$

$$\Rightarrow (4\sqrt{2x-3})^2 = (-x+6)^2$$

$$\Rightarrow 16(2x-3) = (-x+6)^2$$

$$\Rightarrow 16(2x-3) = x^2 - 12x + 36$$

$$\Rightarrow x^2 - 44x + 84 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{44 \pm \sqrt{44^2 - 4 \cdot 84}}{2}$$

$$= \frac{44 \pm 40}{2} = 22 \pm 20$$

$$\begin{matrix} x = 2 \\ \text{OR} \\ 42 \end{matrix}$$

$$\begin{matrix} x = 32 \\ \text{OR} \\ 12 \end{matrix}$$